

respirometry

```
# PreSens SDR respirometry workflow
# OXYGEN-CONCENTRATION VERSION
# Multi-run version using:
# 1) real clock-time windows
# 2) manual well assignment
# 3) manual discard wells
# 4) pedal disk area normalization
# 5) oxygen converted from % air saturation to umol/L
```

```
library(readr)
library(readxl)
library(dplyr)
library(tidyr)
library(stringr)
library(purrr)
library(ggplot2)
library(lubridate)
library(broom)
library(car)
library(emmeans)
library(multcomp)
library(multcompView)
library(scales)
library(respirometry)
```

```
rm(list = ls())
#graphics.off()

getwd()
```

```
## [1] "/Users/junbc/Downloads"
```

```
select <- dplyr::select
filter <- dplyr::filter
mutate <- dplyr::mutate
lag <- dplyr::lag
```

```
## 1.1 Working directory + output
```

```
setwd("/Users/junbc/Documents/GitHub/heating_lacerates_final")
```

```
out_dir <- "~/Documents/GitHub/heating_lacerates_final/data/respirometry/processed_outputs_respirometry"
dir.create(
  out_dir,
```

```

recursive = TRUE,
showWarnings = FALSE
)

## 1.2 Chamber volume
chamber_volume_l <- 80e-6

# 2) INPUT DATA

## 2.1 Run metadata
run_info <- tribble(
  ~plate_name, ~file, ~run_date, ~tissue_type, ~temperature, ~date_modified, ~start_clock, ~win_start,

  "20260326_Lacerate_25C_Plate4",
  "SDR 1106_032626_65531 PM_lacerate_sym_apo_plate4_25C_03.26.26.csv",
  "03/26/26", "Lacerate", "25C", "03/27/26 2:08 PM", "6:58:14 PM", "20:30", "23:30",
  "D3, D6, B4, C3, C4",
  "A1, A3, D2, D4",
  "C1, C2, C5, C6, D1, D5",
  "A2, A4, A5, A6, B1, B2, B3, B5, B6",

  "20260324_Lacerate_32C_Plate3",
  "SDR 1106_032426_54943 PM_lacerate_sym_apo_plate3_32C_03.24.26.csv",
  "03/24/26", "Lacerate", "32C", "03/25/26 3:53 PM", "5:53:09 PM", "19:15", "22:15",
  "D2, D3, D6, C1, C6",
  "A2, A3, D5",
  "C2, C3, C4, C5, D1, D4",
  "A1, A4, A5, A6, B1, B2, B3, B4, B5, B6",

  "20260319_Lacerate_25C_Plate2",
  "SDR 1106_031926_24601 PM_lacerate_sym_apo_25C_03.19.26.csv",
  "03/19/26", "Lacerate", "25C", "03/20/26 12:54 PM", "2:49:47 PM", "16:30", "19:00",
  "B2, A6",
  "B1, B6, D1, D3",
  "C1, C2, C3, C4, C5, C6, D2, D4, D5, D6",
  "A1, A2, A3, A4, A5, B3, B4, B5",

  "20260317_Lacerate_32C_Plate1",
  "SDR 1106_031726_21405 PM_lacerate_sym_apo_32C_03.17.26.csv",
  "03/17/26", "Lacerate", "32C", "03/18/26 10:41 AM", "2:16:43 PM", "15:15", "18:15",
  "D3",
  "A1, A6, D1, D6",
  "C1, C2, C3, C4, C5, C6, D2, D4, D5",
  "A2, A3, A4, A5, B1, B2, B3, B4, B5, B6"
)

## 2.2 Area data
area_data <- read_csv("~/Documents/GitHub/heating_lacerates_final/data/respirometry_area_data_raw.csv")
mutate(
  area = area / 100,
  side_estimated = str_detect(str_to_lower(notes), "on side"),
  note_rank = ifelse(side_estimated, 1, 0)
) %>%

```

```

arrange(plate_name, well, note_rank) %>%
group_by(plate_name, well) %>%
slice(1) %>%
ungroup() %>%
select(label, area, notes, well, plate_name)

split_wells <- function(x) {
  if (is.na(x) || x == "") return(character(0))
  str_split(x, ",")[[1]] |> str_trim()
}

time_to_sec <- function(x) {
  parts <- str_split(x, ":", simplify = TRUE)
  as.numeric(parts[, 1]) * 3600 +
  as.numeric(parts[, 2]) * 60 +
  as.numeric(parts[, 3])
}

get_settings <- function(file) {
  lines <- read_lines(file, locale = locale(encoding = "UTF-16"))

  get_val <- function(label) {
    hit <- lines[str_detect(lines, fixed(label))][1]
    if (is.na(hit)) return(NA_character_)

    parts <- str_split(hit, "\\t", simplify = TRUE)
    parts <- str_trim(parts)
    parts <- parts[parts != ""]

    if (length(parts) >= 2) return(parts[length(parts)])

    hit |> str_remove(fixed(label)) |> str_trim()
  }

  tibble(
    barometric_pressure_hpa = as.numeric(get_val("Barometric pressure [hPa]:")),
    temperature_c_file = as.numeric(get_val("Temperature [°C]:")),
    salinity_ppt_file = as.numeric(get_val("Salinity [%]:")),
    chamber_volume_ul_file = as.numeric(get_val("Chamber volume [µL]:")),
    oxygen_unit_file = get_val("Oxygen unit:")
  )
}

fit_slope <- function(df) {
  df <- df %>% arrange(time_s)

  if (nrow(df) < 2) {
    return(tibble(
      n_points = nrow(df),
      slope_o2_umol_l_per_s = NA_real_,
      slope_o2_umol_l_per_h = NA_real_,
      r2 = NA_real_,

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      start_time = as.POSIXct(NA),
      end_time = as.POSIXct(NA)
    ))
  }

mod <- lm(oxygen_conc_umol_l ~ time_s, data = df)

tibble(
  n_points = nrow(df),
  slope_o2_umol_l_per_s = coef(mod)[["time_s"]],
  slope_o2_umol_l_per_h = coef(mod)[["time_s"]] * 3600,
  r2 = summary(mod)$r.squared,
  start_time = min(df$clock_time, na.rm = TRUE),
  end_time = max(df$clock_time, na.rm = TRUE)
)
}

# 4) STORAGE OBJECTS

all_outputs <- vector("list", nrow(run_info))
names(all_outputs) <- run_info$plate_name

all_settings_used <- list()
all_results <- list()
all_rates_flagged <- list()
all_manual_plate_keys <- list()
all_area_used <- list()
all_rates_with_area <- list()
all_rates_bio <- list()
all_missing_area <- list()
all_dat_long <- list()

# 5) RUN EACH PLATE

for (i in seq_len(nrow(run_info))) {

  meta <- run_info[i, , drop = FALSE]
  this_plate_name <- meta$plate_name[[1]]
  file <- file.path(
    path.expand("~/Documents/GitHub/heating_lacerates_final/data/respirometry"),
    meta$file[[1]]
  )
  message("Processing: ", this_plate_name)

  # 5.1 Settings + raw data

  settings <- get_settings(file)

  lines <- read_lines(file, locale = locale(encoding = "UTF-16"))
  header_line <- which(str_detect(lines, "^Date \\[DD-MM-YYYY\\]"))[1]

  raw_dat <- read_tsv(
    I(paste(lines[header_line:length(lines)], collapse = "\n")),

```

```

  show_col_types = FALSE
)

oxygen_cols <- names(raw_dat)[str_detect(names(raw_dat), "\\[Oxygen\\]")]

# 5.2 Time + window

start_datetime <- mdy_hms(paste(meta$run_date[[1]], meta$start_clock[[1]]))
window_start   <- mdy_hm(paste(meta$run_date[[1]], meta$win_start[[1]]))
window_end     <- mdy_hm(paste(meta$run_date[[1]], meta$win_end[[1]]))

# 5.3 Manual well groups

manual_plate_key <- bind_rows(
  tibble(well = split_wells(meta$discard[[1]]), manual_group = "Discard"),
  tibble(well = split_wells(meta$real_blanks[[1]]), manual_group = "Blank"),
  tibble(well = split_wells(meta$apo[[1]]), manual_group = "Apo"),
  tibble(well = split_wells(meta$sym[[1]]), manual_group = "Sym")
) %>%
  distinct(well, .keep_all = TRUE) %>%
  mutate(plate_name = this_plate_name)

# 5.4 Long format + oxygen conversion

dat_long_raw <- raw_dat %>%
  select(`Relative time [HH:MM:SS]`, all_of(oxygen_cols)) %>%
  mutate(
    time_s = time_to_sec(`Relative time [HH:MM:SS]`),
    clock_time = start_datetime + seconds(time_s)
  ) %>%
  pivot_longer(
    cols = -c(`Relative time [HH:MM:SS]`, time_s, clock_time),
    names_to = "well",
    values_to = "oxygen_percent_air_sat"
  ) %>%
  mutate(
    well = str_remove(well, " \\[Oxygen\\]"),
    oxygen_percent_air_sat = as.numeric(oxygen_percent_air_sat)
  ) %>%
  left_join(
    manual_plate_key %>% select(well, manual_group),
    by = "well"
  ) %>%
  mutate(
    oxygen_conc_umol_l = respirometry::conv_o2(
      o2 = oxygen_percent_air_sat,
      from = "percent_a.s.",
      to = "umol_per_l",
      temp = settings$temperature_c_file,
      sal = settings$salinity_ppt_file,
      atm_pres = settings$barometric_pressure_hpa / 1013.25
    ),
    plate_name = this_plate_name,

```

```

    temperature = meta$temperature[[1]]
  )

dat_use <- dat_long_raw %>%
  filter(
    clock_time >= window_start,
    clock_time <= window_end
  )

dat_analysis <- dat_use %>%
  filter(
    !is.na(manual_group),
    manual_group != "Discard"
  )

# 5.5 Fit slopes

results <- map_dfr(unique(dat_analysis$well), function(w) {
  this <- dat_analysis %>% filter(well == w)

  bind_cols(
    tibble(
      plate_name = this_plate_name,
      well = w,
      treatment = unique(this$manual_group)
    ),
    fit_slope(this)
  )
}) %>%
  mutate(
    oxygen_decline_rate_umol_l_h = -slope_o2_umol_l_per_h
  )

# 5.6 Blank threshold + QC flags

blank_mean <- mean(
  results$oxygen_decline_rate_umol_l_h[results$treatment == "Blank"],
  na.rm = TRUE
)

blank_sd <- sd(
  results$oxygen_decline_rate_umol_l_h[results$treatment == "Blank"],
  na.rm = TRUE
)

threshold <- blank_mean + 3 * blank_sd

rates_flagged <- results %>%
  mutate(
    blank_threshold = threshold,
    is_biological = oxygen_decline_rate_umol_l_h > threshold,
    poor_fit = r2 < 0.9,
    unreliable = !is_biological | poor_fit
  )

```

```

)

# 5.7 Join area

area_this_plate <- area_data %>%
  filter(plate_name == this_plate_name)

rates_with_area <- rates_flagged %>%
  left_join(area_this_plate, by = c("plate_name", "well")) %>%
  mutate(
    respiration_norm = oxygen_decline_rate_umol_l_h / area
  )

rates_bio <- rates_with_area %>%
  filter(treatment %in% c("Apo", "Sym"))

missing_area <- rates_with_area %>%
  filter(is.na(area))

all_outputs[[i]] <- list(
  settings = settings %>% mutate(plate_name = this_plate_name, file = file),
  manual_plate_key = manual_plate_key,
  dat_long = dat_long_raw,
  dat_use = dat_use,
  dat_analysis = dat_analysis,
  results = results,
  rates_flagged = rates_flagged,
  area_this_plate = area_this_plate,
  rates_with_area = rates_with_area,
  rates_bio = rates_bio,
  missing_area = missing_area
)

all_settings_used[[this_plate_name]] <- all_outputs[[i]]$settings
all_results[[this_plate_name]] <- all_outputs[[i]]$results
all_rates_flagged[[this_plate_name]] <- all_outputs[[i]]$rates_flagged
all_manual_plate_keys[[this_plate_name]] <- all_outputs[[i]]$manual_plate_key
all_area_used[[this_plate_name]] <- all_outputs[[i]]$area_this_plate
all_rates_with_area[[this_plate_name]] <- all_outputs[[i]]$rates_with_area
all_rates_bio[[this_plate_name]] <- all_outputs[[i]]$rates_bio
all_missing_area[[this_plate_name]] <- all_outputs[[i]]$missing_area
all_dat_long[[this_plate_name]] <- all_outputs[[i]]$dat_long

write_csv(
  rates_with_area,
  file.path(out_dir, paste0(this_plate_name, "_results.csv"))
)
}

```

6) COMBINE OUTPUTS

```
all_settings_used <- bind_rows(all_settings_used)
all_results <- bind_rows(all_results)
all_rates_flagged <- bind_rows(all_rates_flagged)
all_manual_plate_keys <- bind_rows(all_manual_plate_keys)
all_area_used <- bind_rows(all_area_used)
all_rates_with_area <- bind_rows(all_rates_with_area)
all_rates_bio <- bind_rows(all_rates_bio)
all_missing_area <- bind_rows(all_missing_area)
all_dat_long <- bind_rows(all_dat_long)
```

optional alias if you used this old name anywhere

```
all_rates <- all_results
```

```
print(all_rates_bio)
```

```
## # A tibble: 68 x 18
##   plate_name      well treatment n_points slope_o2_umol_l_per_s
##   <chr>          <chr> <chr>      <int>          <dbl>
## 1 20260326_Lacerate_25C_Plate4 A2    Sym        36      -0.00000467
## 2 20260326_Lacerate_25C_Plate4 A4    Sym        36      -0.00000220
## 3 20260326_Lacerate_25C_Plate4 A5    Sym        36      -0.00000396
## 4 20260326_Lacerate_25C_Plate4 A6    Sym        36      -0.00000281
## 5 20260326_Lacerate_25C_Plate4 B1    Sym        36      -0.00000236
## 6 20260326_Lacerate_25C_Plate4 B2    Sym        36      -0.00000135
## 7 20260326_Lacerate_25C_Plate4 B3    Sym        36      -0.00000229
## 8 20260326_Lacerate_25C_Plate4 B5    Sym        36      -0.00000262
## 9 20260326_Lacerate_25C_Plate4 B6    Sym        36      -0.00000175
## 10 20260326_Lacerate_25C_Plate4 C1    Apo        36      -0.00000149
## # i 58 more rows
## # i 13 more variables: slope_o2_umol_l_per_h <dbl>, r2 <dbl>,
## #   start_time <dtm>, end_time <dtm>, oxygen_decline_rate_umol_l_h <dbl>,
## #   blank_threshold <dbl>, is_biological <lgl>, poor_fit <lgl>,
## #   unreliable <lgl>, label <chr>, area <dbl>, notes <chr>,
## #   respiration_norm <dbl>
```

```
all_rates_bio %>%
  select(
    plate_name,
    well,
    treatment,
    slope_o2_umol_l_per_h,
    oxygen_decline_rate_umol_l_h,
    area,
    respiration_norm,
    r2,
    unreliable,
    notes
  ) %>%
  arrange(plate_name, treatment, well)
```



```
## # A tibble: 68 x 10
##   plate_name      well treatment slope_o2_umol_l_per_h oxygen_decline_rate_~1
##   <chr>          <chr> <chr>          <dbl>          <dbl>
## 1 20260317_Lacera~ C1    Apo           -0.0119         0.0119
## 2 20260317_Lacera~ C2    Apo           -0.0192         0.0192
## 3 20260317_Lacera~ C3    Apo           -0.0122         0.0122
## 4 20260317_Lacera~ C4    Apo           -0.0243         0.0243
## 5 20260317_Lacera~ C5    Apo           -0.0139         0.0139
## 6 20260317_Lacera~ C6    Apo           -0.0149         0.0149
## 7 20260317_Lacera~ D2    Apo           -0.00959        0.00959
## 8 20260317_Lacera~ D4    Apo           -0.0309         0.0309
## 9 20260317_Lacera~ D5    Apo           -0.0295         0.0295
## 10 20260317_Lacera~ A2    Sym           -0.0392         0.0392
## # i 58 more rows
## # i abbreviated name: 1: oxygen_decline_rate_umol_l_h
## # i 5 more variables: area <dbl>, respiration_norm <dbl>, r2 <dbl>,
## #   unreliable <lgl>, notes <chr>
```

```
all_missing_area
```

```
## # A tibble: 11 x 18
##   plate_name      well treatment n_points slope_o2_umol_l_per_s
##   <chr>          <chr> <chr>      <int>          <dbl>
## 1 20260326_Lacera~ 25C_Plate4 D2    Blank         36          -6.42e-8
## 2 20260326_Lacera~ 25C_Plate4 D4    Blank         36          -8.74e-8
## 3 20260324_Lacera~ 32C_Plate3 D5    Blank         36          -7.66e-7
## 4 20260319_Lacera~ 25C_Plate2 B1    Blank         30          -8.76e-8
## 5 20260319_Lacera~ 25C_Plate2 B6    Blank         30          -3.14e-9
## 6 20260319_Lacera~ 25C_Plate2 D1    Blank         30          -9.58e-8
## 7 20260319_Lacera~ 25C_Plate2 D3    Blank         30          -1.63e-8
## 8 20260317_Lacera~ 32C_Plate1 A1    Blank         36          -4.81e-7
## 9 20260317_Lacera~ 32C_Plate1 A6    Blank         36          -1.36e-6
## 10 20260317_Lacera~ 32C_Plate1 D1    Blank         36          -4.96e-7
## 11 20260317_Lacera~ 32C_Plate1 D6    Blank         36          -8.12e-7
## # i 13 more variables: slope_o2_umol_l_per_h <dbl>, r2 <dbl>,
## #   start_time <dtm>, end_time <dtm>, oxygen_decline_rate_umol_l_h <dbl>,
## #   blank_threshold <dbl>, is_biological <lgl>, poor_fit <lgl>,
## #   unreliable <lgl>, label <chr>, area <dbl>, notes <chr>,
## #   respiration_norm <dbl>
```

```
# 13) Full trace plots with captured window overlaid
```

```
all_full_traces <- purrr::map_dfr(all_outputs, "dat_long") %>%
  filter(!is.na(oxygen_conc_umol_l)) %>%
  mutate(
    well = as.character(well),
    oxygen_conc_nmol_l = oxygen_conc_umol_l * 1000,
    manual_group = case_when(
      is.na(manual_group) ~ "Unassigned",
      TRUE ~ manual_group
    ),
    manual_group = factor(
      manual_group,
```

```

    levels = c("Unassigned", "Discard", "Blank", "Apo", "Sym")
  )
)

window_info <- run_info %>%
  mutate(
    plate_start_clock = lubridate::mdy_hms(paste(run_date, start_clock)),
    window_start_clock = lubridate::mdy_hm(paste(run_date, win_start)),
    window_end_clock = lubridate::mdy_hm(paste(run_date, win_end))
  ) %>%
  select(
    plate_name,
    plate_start_clock,
    window_start_clock,
    window_end_clock
  )

all_full_traces <- all_full_traces %>%
  left_join(window_info, by = "plate_name")

trace_labels <- all_full_traces %>%
  group_by(plate_name, well, manual_group) %>%
  filter(clock_time == max(clock_time, na.rm = TRUE)) %>%
  ungroup()

manual_trace_counts <- all_full_traces %>%
  distinct(plate_name, well, manual_group) %>%
  count(plate_name, manual_group)

write_csv(
  manual_trace_counts,
  file.path(out_dir, "ALL_RUNS_manual_trace_counts_oxygen_concentration_nmol.csv")
)

## Rework for Supplemental Figure to make hours 0 to 24
all_full_traces <- all_full_traces %>%
  group_by(plate_name) %>%
  filter(
    manual_group != "Discard",
  ) %>%
  mutate(
    time_hours = as.numeric(difftime(clock_time, min(clock_time), units = "hours"))
  ) %>%
  ungroup()

window_info_plot <- window_info %>%
  mutate(
    window_start_hours_plot = as.numeric(difftime(window_start_clock, plate_start_clock, units = "hours")),
    window_end_hours_plot = as.numeric(difftime(window_end_clock, plate_start_clock, units = "hours"))
  )

```

```

trace_labels_right <- all_full_traces %>%
  group_by(plate_name, well) %>%
  slice_max(order_by = time_hours, n = 1, with_ties = FALSE) %>%
  ungroup()

FigS3_all_tracing <- ggplot(
  all_full_traces,
  aes(
    x = time_hours,
    y = oxygen_conc_nmol_l,
    group = well,
    color = manual_group
  )
) +
  geom_rect(
    data = window_info_plot,
    aes(
      xmin = window_start_hours_plot,
      xmax = window_end_hours_plot,
      ymin = -Inf,
      ymax = Inf
    ),
    inherit.aes = FALSE,
    fill = "grey70",
    alpha = 0.15
  ) +
  geom_line(alpha = 0.85, linewidth = 1.2) +
  geom_text(
    data = trace_labels_right,
    aes(
      x = time_hours,
      y = oxygen_conc_nmol_l,
      label = well
    ),
    hjust = -0.05,
    size = 2.7,
    show.legend = FALSE
  ) +
  facet_wrap(
    ~ plate_name,
    scales = "free_x",
    ncol = 2,
    labeller = labeller(
      plate_name = c(
        "20260317_Lacerate_32C_Plate1" = "Plate A - 32°C",
        "20260319_Lacerate_25C_Plate2" = "Plate B - 25°C",
        "20260324_Lacerate_32C_Plate3" = "Plate C - 32°C",
        "20260326_Lacerate_25C_Plate4" = "Plate D - 25°C"
      )
    )
  ) +
  scale_color_manual(
    values = c(

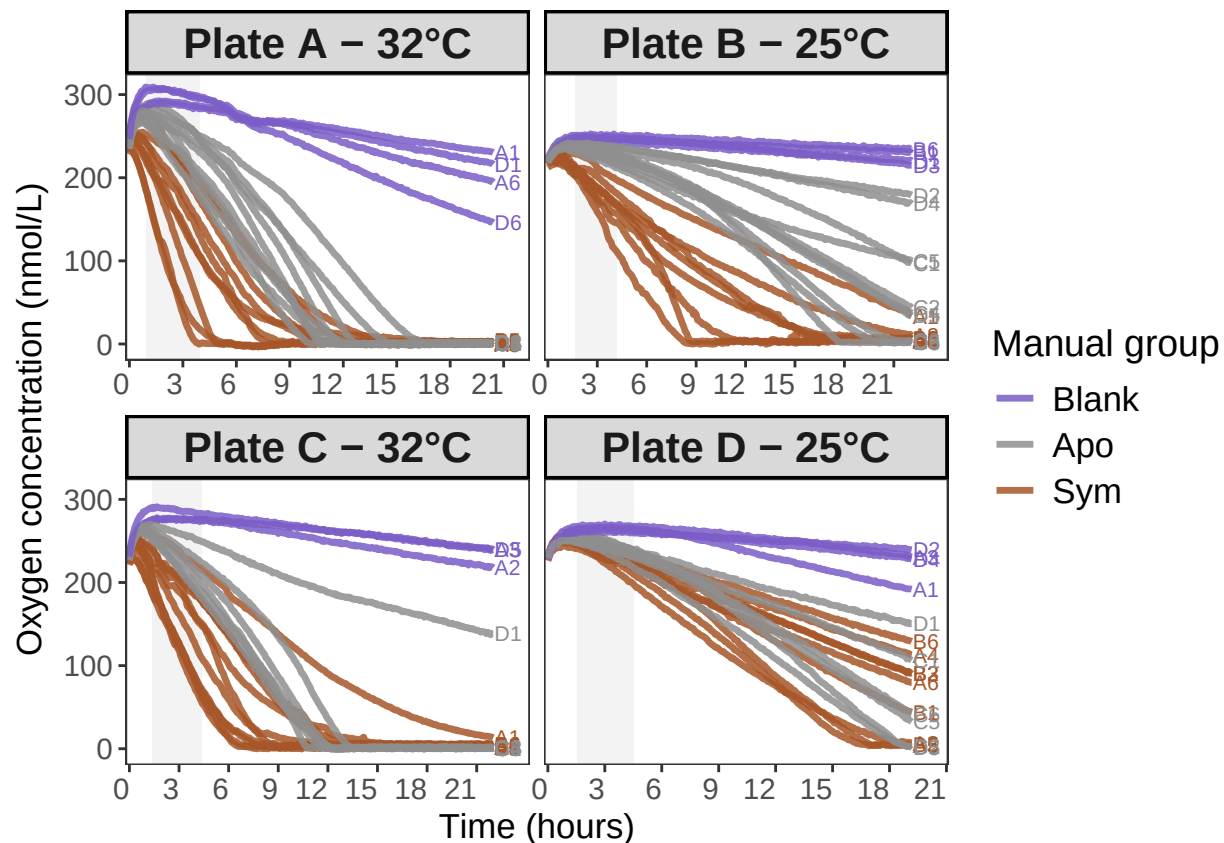
```

```

    "Unassigned" = "#D9D9D9",
#    "Discard"    = "#E66101",
    "Blank"      = "#7A5DC7",
    "Apo"        = "#8F8F8F",
    "Sym"        = "#A65628"
  ),
  na.value = "#EAEAEA"
) +
scale_x_continuous(
  breaks = seq(
    0,
    ceiling(max(all_full_traces$time_hours, na.rm = TRUE)),
    by = 3
  ),
  expand = expansion(mult = c(0.01, 0.10))
) +
labs(
  x = "Time (hours)",
  y = expression(paste("Oxygen concentration (nmol/L)")),
  color = "Manual group"
) +
theme_bw() +
theme(
  text = element_text(family = "sans"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "white", color = NA),
  plot.background = element_rect(fill = "white", color = NA),
  strip.background = element_rect(fill = "grey85", color = "black", linewidth = 1),
  strip.text = element_text(size = 16, face = "bold"),
  axis.title = element_text(size = 13),
  axis.text = element_text(size = 11),
  axis.text.x = element_text(angle = 0, hjust = 1),
  legend.title = element_text(size = 14),
  legend.text = element_text(size = 13),
  legend.key.size = unit(1.2, "lines"),
  legend.position = "right"
)

print(FigS3_all_tracing)

```



```

ggsave(
  filename = "FigS3_ALL_RUNS_full_trace_manual_groups_labeled_oxygen_concentration_nmol_tight.png",
  plot = FigS3_all_tracing,
  path = "~/Documents/GitHub/heating_lacerates_final/figs",
  device = "png",
  width = 15,
  height = 9,
  units = "in",
  dpi = 600,
  bg = "white"
)

ggsave(
  filename = "FigS3_ALL_RUNS_full_trace_manual_groups_labeled_oxygen_concentration_nmol_tight.pdf",
  plot = FigS3_all_tracing,
  path = "~/Documents/GitHub/heating_lacerates_final/figs",
  device = pdf,
  width = 15,
  height = 9,
  units = "in",
  bg = "white"
)

# 14) Pedal disc size plot
area_plot_dat <- area_data %>%
  left_join(

```

```

    all_manual_plate_keys %>% select(plate_name, well, manual_group),
    by = c("plate_name", "well")
  ) %>%
  filter(manual_group %in% c("Apo", "Sym")) %>%
  mutate(
    temperature = case_when(
      str_detect(plate_name, "25C") ~ "25C",
      str_detect(plate_name, "32C") ~ "32C"
    ),
    treatment = factor(manual_group, levels = c("Apo", "Sym")),
    temperature = factor(temperature, levels = c("25C", "32C"))
  )

p_pedal_area <- ggplot(
  area_plot_dat %>%
  mutate(
    color_group = case_when(
      treatment == "Sym" & temperature == "25C" ~ "sym_25",
      treatment == "Apo" & temperature == "25C" ~ "apo_25",
      treatment == "Sym" & temperature == "32C" ~ "sym_32",
      treatment == "Apo" & temperature == "32C" ~ "apo_32"
    )
  ),
  aes(
    x = temperature,
    y = area,
    color = color_group
  )
) +
  geom_boxplot(
    width = 0.42,
    fill = NA,
    linewidth = 1.4,
    outlier.shape = NA,
    show.legend = FALSE
  ) +
  geom_jitter(
    width = 0.08,
    size = 2.0,
    alpha = 0.7,
    show.legend = FALSE
  ) +
  facet_wrap(~ factor(treatment, levels = c("Sym", "Apo"))) +

  scale_color_manual(
    values = c(
      "sym_25" = "#3B6FB6",
      "apo_25" = "#6FA3D9",
      "sym_32" = "#E64B35",
      "apo_32" = "#F39B7F"
    )
  )
) +

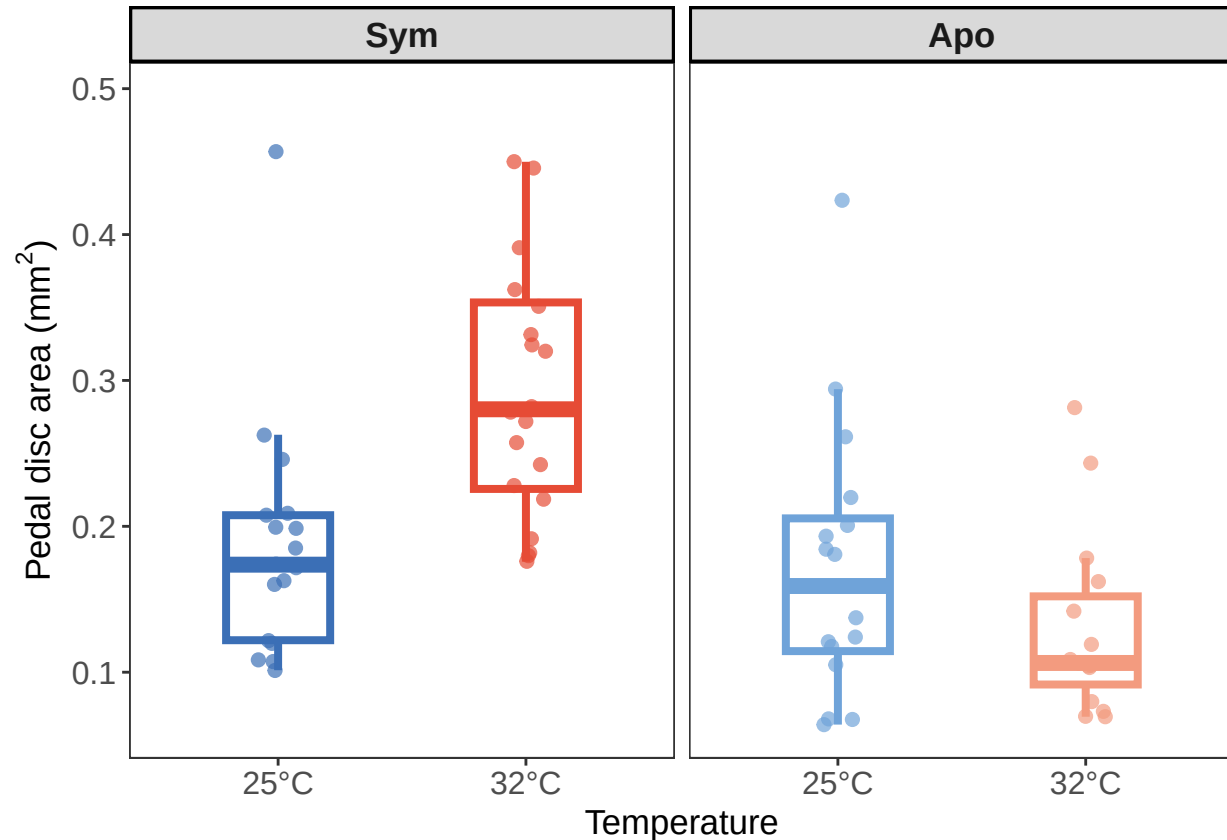
```

```

scale_x_discrete(
  labels = c("25C" = "25°C", "32C" = "32°C")
) +
scale_y_continuous(
  expand = expansion(mult = c(0.02, 0.04))
) +
coord_cartesian(ylim = c(0.05, 0.5)) +
labs(
  x = "Temperature",
  y = expression(paste("Pedal disc area (mm"2, ")"))
) +
theme_bw() +
theme(
  text = element_text(family = "sans"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "white", color = NA),
  plot.background = element_rect(fill = "white", color = NA),
  strip.background = element_rect(fill = "grey85", color = "black", linewidth = 1),
  strip.text = element_text(size = 13, face = "bold"),
  axis.title = element_text(size = 13),
  axis.text = element_text(size = 12),
  legend.position = "none"
)

print(p_pedal_area)

```



```

ggsave(
  filename = "FigS2_Resp_pedal_area.png",
  plot = p_pedal_area,
  path = "~/Documents/GitHub/heating_lacerates_final/figs",
  device = "png",
  width = 7,
  height = 5,
  units = "in",
  dpi = 600,
  bg = "white"
)

ggsave(
  filename = "FigS2_Resp_pedal_area.pdf",
  plot = p_pedal_area,
  path = "~/Documents/GitHub/heating_lacerates_final/figs",
  device = pdf,
  width = 7,
  height = 5,
  units = "in",
  bg = "white"
)

# 15) Temperature x treatment model on normalized oxygen respiration

stats_dat2 <- all_rates_bio %>%
  filter(
    treatment %in% c("Sym", "Apo"),
    !is.na(oxygen_decline_rate_umol_l_h),
    !is.na(respiration_norm)
  ) %>%
  mutate(
    oxygen_decline_rate_nmol_l_h = oxygen_decline_rate_umol_l_h * 1000,
    respiration_norm_nmol = respiration_norm * 1000,
    temperature = case_when(
      str_detect(plate_name, "25C") ~ "25C",
      str_detect(plate_name, "32C") ~ "32C"
    ),
    temperature = factor(temperature, levels = c("25C", "32C")),
    treatment = factor(treatment, levels = c("Apo", "Sym")),
    plate_name = factor(plate_name)
  )

mod <- lm(respiration_norm_nmol ~ treatment * temperature, data = stats_dat2)

summary(mod)

```

```

##
## Call:
## lm(formula = respiration_norm_nmol ~ treatment * temperature,
##     data = stats_dat2)
##
## Residuals:

```

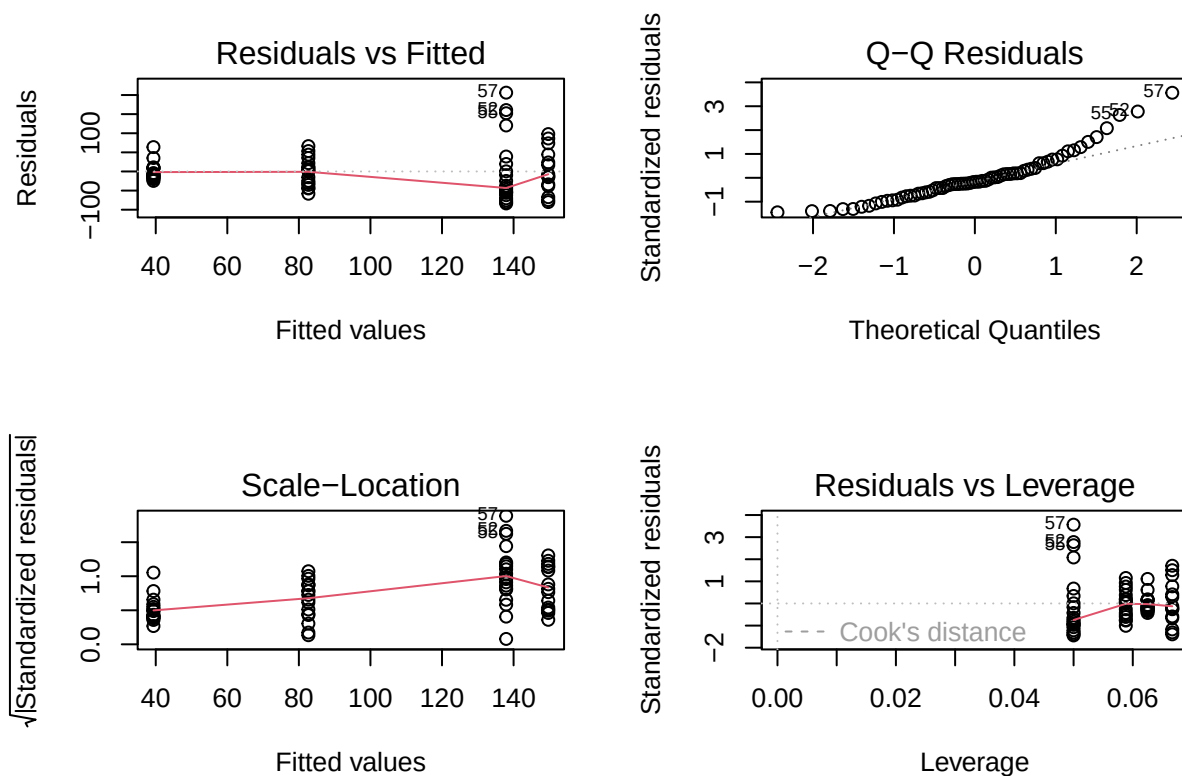


```
##      Min      1Q Median      3Q      Max
## -83.65 -37.66  -9.89  20.17 206.17
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      39.36      14.84   2.653  0.0100 *
## treatmentSym      43.30      20.67   2.095  0.0402 *
## temperature32C    110.33      21.33   5.173 2.48e-06 ***
## treatmentSym:temperature32C -55.07      28.95  -1.902  0.0617 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 59.35 on 64 degrees of freedom
## Multiple R-squared:  0.3638, Adjusted R-squared:  0.334
## F-statistic: 12.2 on 3 and 64 DF, p-value: 2.073e-06
```

```
anova(mod)
```

```
## Analysis of Variance Table
##
## Response: respiration_norm_nmol
##              Df Sum Sq Mean Sq F value    Pr(>F)
## treatment      1   6605    6605   1.8751  0.17568
## temperature     1 109565  109565  31.1053 5.277e-07 ***
## treatment:temperature 1  12741   12741   3.6173  0.06169 .
## Residuals      64 225433    3522
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
par(mfrow = c(2, 2))
plot(mod)
```



```
par(mfrow = c(1, 1))

shapiro.test(residuals(mod))

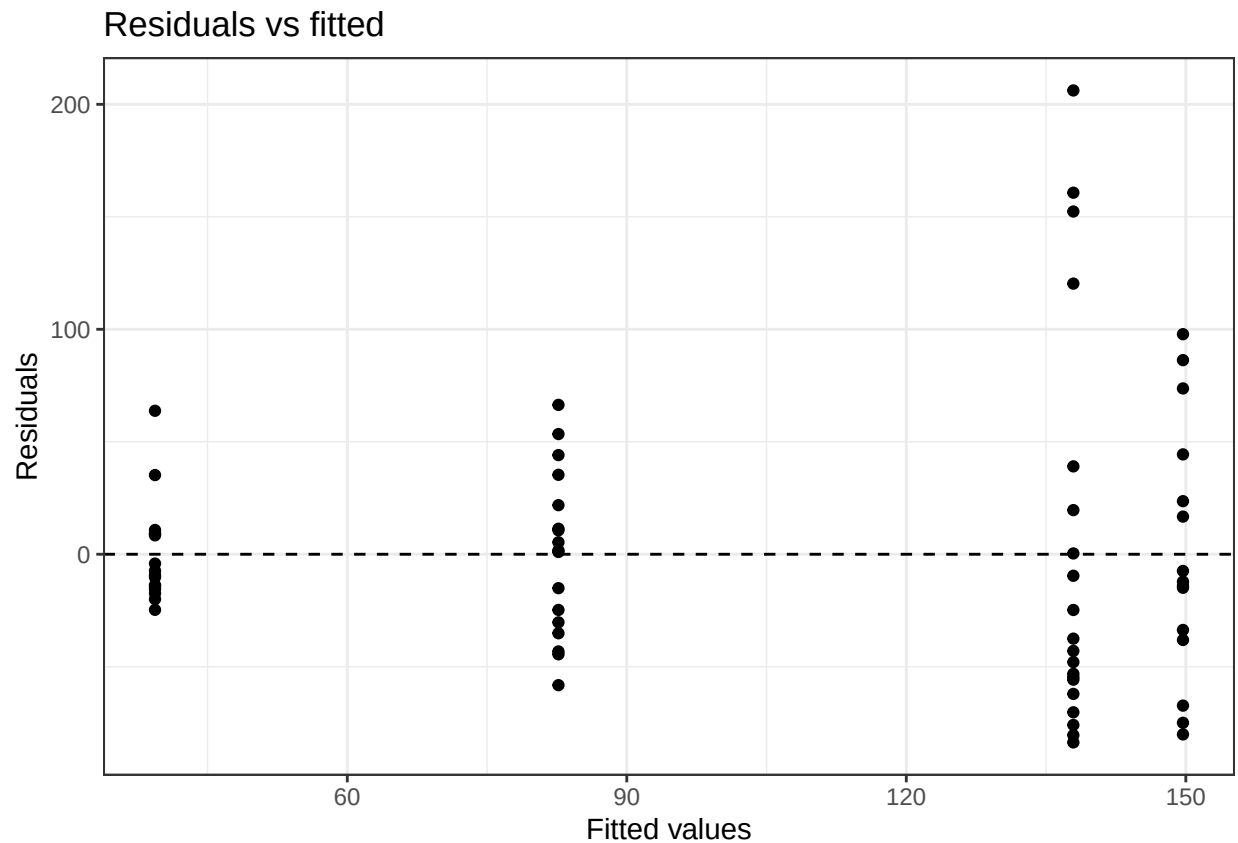
##
##  Shapiro-Wilk normality test
##
## data:  residuals(mod)
## W = 0.90776, p-value = 9.996e-05

car::leveneTest(respiration_norm_nmol ~ treatment * temperature, data = stats_dat2)

## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group 3  3.4532 0.02154 *
##      64
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

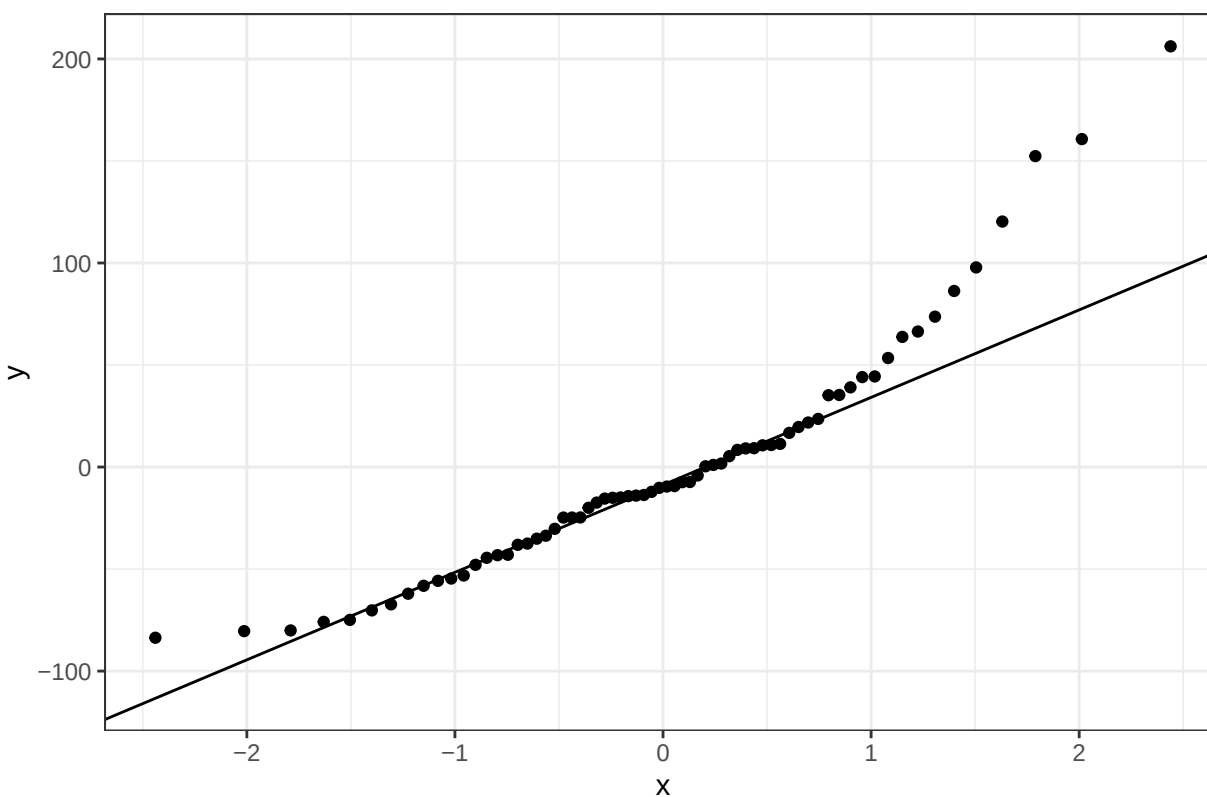
diag_dat <- data.frame(
  fitted = fitted(mod),
  resid = residuals(mod)
)
```

```
ggplot(diag_dat, aes(fitted, resid)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = 2) +
  theme_bw() +
  labs(
    title = "Residuals vs fitted",
    x = "Fitted values",
    y = "Residuals"
  )
)
```



```
ggplot(diag_dat, aes(sample = resid)) +
  stat_qq() +
  stat_qq_line() +
  theme_bw() +
  labs(title = "QQ plot of residuals")
)
```

QQ plot of residuals



```
emmeans(mod, pairwise ~ treatment | temperature)
```

```
## $emmeans
## temperature = 25C:
##   treatment emmean   SE df lower.CL upper.CL
## Apo         39.4 14.8 64    9.72     69
## Sym         82.7 14.4 64   53.91    111
##
## temperature = 32C:
##   treatment emmean   SE df lower.CL upper.CL
## Apo        149.7 15.3 64   119.08    180
## Sym        137.9 13.3 64   111.42    164
##
## Confidence level used: 0.95
##
## $contrasts
## temperature = 25C:
##   contrast estimate   SE df t.ratio p.value
## Apo - Sym   -43.3 20.7 64   -2.095  0.0402
##
## temperature = 32C:
##   contrast estimate   SE df t.ratio p.value
## Apo - Sym    11.8 20.3 64    0.580  0.5637
```

```
emmeans(mod, pairwise ~ temperature | treatment)
```

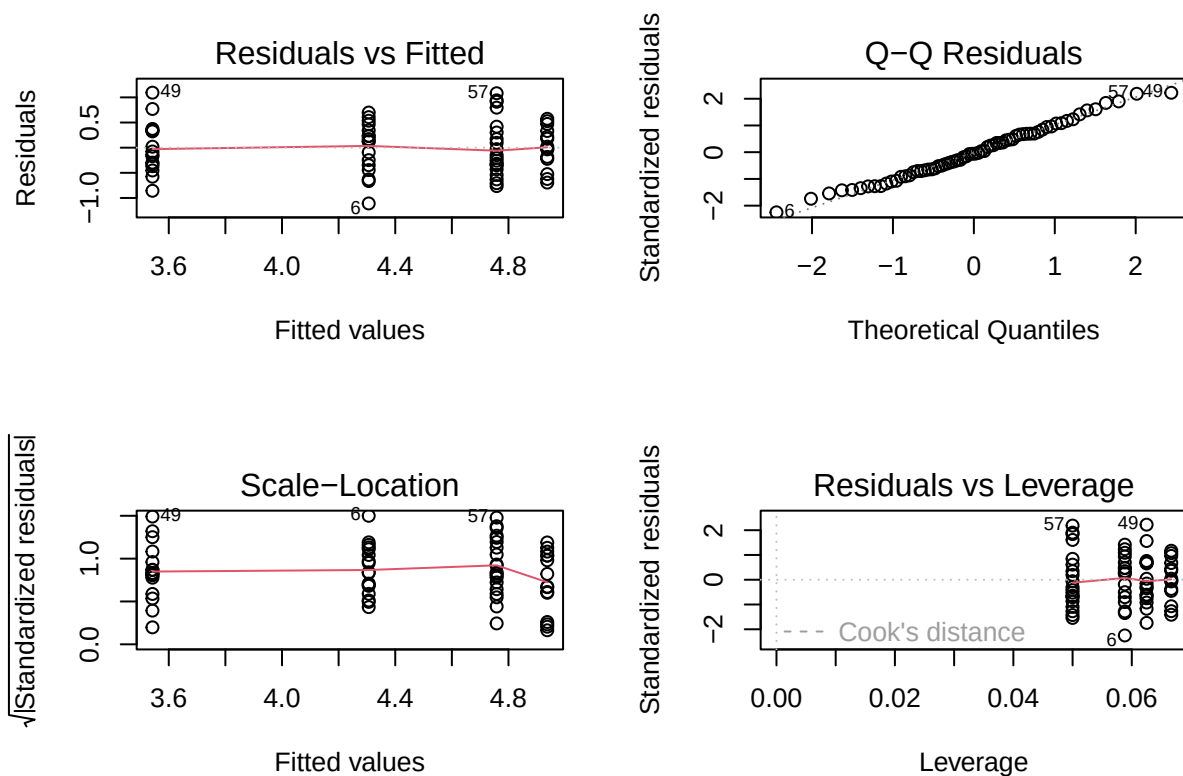
```
## $emmeans
## treatment = Apo:
##   temperature emmean    SE df lower.CL upper.CL
##   25C          39.4 14.8 64     9.72     69
##   32C          149.7 15.3 64    119.08    180
##
## treatment = Sym:
##   temperature emmean    SE df lower.CL upper.CL
##   25C          82.7 14.4 64     53.91    111
##   32C          137.9 13.3 64    111.42    164
##
## Confidence level used: 0.95
##
## $contrasts
## treatment = Apo:
##   contrast estimate    SE df t.ratio p.value
##   25C - 32C   -110.3 21.3 64    -5.173  <.0001
##
## treatment = Sym:
##   contrast estimate    SE df t.ratio p.value
##   25C - 32C    -55.3 19.6 64    -2.823  0.0063
```

```
## Logtransformed model
```

```
mod_log <- lm(log(respiration_norm_nmol) ~ treatment * temperature, data = stats_dat2)
```

```
par(mfrow = c(2, 2))
```

```
plot(mod_log)
```



```
par(mfrow = c(1, 1))

shapiro.test(residuals(mod_log))

##
## Shapiro-Wilk normality test
##
## data: residuals(mod_log)
## W = 0.98765, p-value = 0.741

car::leveneTest(log(respiration_norm_nmol) ~ treatment * temperature, data = stats_dat2)

## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group 3  0.563 0.6414
##      64

summary(mod_log)

##
## Call:
## lm(formula = log(respiration_norm_nmol) ~ treatment * temperature,
##     data = stats_dat2)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.10879 -0.34809 -0.02806  0.33956  1.09371
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.5424     0.1270  27.886 < 2e-16 ***
## treatmentSym      0.7639     0.1770   4.316 5.62e-05 ***
## temperature32C    1.3946     0.1826   7.637 1.41e-10 ***
## treatmentSym:temperature32C -0.9427     0.2479  -3.803 0.000322 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5081 on 64 degrees of freedom
## Multiple R-squared:  0.5323, Adjusted R-squared:  0.5104
## F-statistic: 24.28 on 3 and 64 DF,  p-value: 1.308e-10
```

```
car::Anova(mod_log, type = "II")
```

```
## Anova Table (Type II tests)
##
## Response: log(respiration_norm_nmol)
##              Sum Sq Df F value    Pr(>F)
## treatment      1.3496  1   5.227 0.0255578 *
## temperature    13.2016  1  51.131 1.012e-09 ***
## treatment:temperature  3.7339  1  14.462 0.0003218 ***
## Residuals      16.5241 64
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(mod_log, pairwise ~ treatment | temperature)
```

```
## $emmeans
## temperature = 25C:
##   treatment emmean    SE df lower.CL upper.CL
## Apo         3.54 0.127 64     3.29     3.80
## Sym         4.31 0.123 64     4.06     4.55
##
## temperature = 32C:
##   treatment emmean    SE df lower.CL upper.CL
## Apo         4.94 0.131 64     4.67     5.20
## Sym         4.76 0.114 64     4.53     4.99
##
## Results are given on the log (not the response) scale.
## Confidence level used: 0.95
##
## $contrasts
## temperature = 25C:
##   contrast estimate    SE df t.ratio p.value
## Apo - Sym   -0.764 0.177 64   -4.316 0.0001
##
## temperature = 32C:
##   contrast estimate    SE df t.ratio p.value
```

```
## Apo - Sym    0.179 0.174 64    1.030 0.3068
##
## Results are given on the log (not the response) scale.
```

```
emmeans(mod_log, pairwise ~ temperature | treatment)
```

```
## $emmeans
## treatment = Apo:
##   temperature emmean      SE df lower.CL upper.CL
##   25C          3.54 0.127 64     3.29     3.80
##   32C          4.94 0.131 64     4.67     5.20
##
## treatment = Sym:
##   temperature emmean      SE df lower.CL upper.CL
##   25C          4.31 0.123 64     4.06     4.55
##   32C          4.76 0.114 64     4.53     4.99
##
## Results are given on the log (not the response) scale.
## Confidence level used: 0.95
##
## $contrasts
## treatment = Apo:
##   contrast estimate      SE df t.ratio p.value
##   25C - 32C   -1.395 0.183 64   -7.637 <.0001
##
## treatment = Sym:
##   contrast estimate      SE df t.ratio p.value
##   25C - 32C   -0.452 0.168 64   -2.696 0.0089
##
## Results are given on the log (not the response) scale.
```

```
emm_log_treat <- emmeans(mod_log, ~ treatment | temperature)

emm_log_treat_resp <- emmeans(mod_log, ~ treatment | temperature, type = "response")

emm_log_temp_resp <- emmeans(mod_log, ~ temperature | treatment, type = "response")

# Table S8. Type II ANOVA for respiration rate

anova_respiration <- car::Anova(mod_log, type = "II")
anova_respiration
```

```
## Anova Table (Type II tests)
##
## Response: log(respiration_norm_nmol)
##              Sum Sq Df F value    Pr(>F)
## treatment          1.3496  1    5.227 0.0255578 *
## temperature       13.2016  1   51.131 1.012e-09 ***
## treatment:temperature  3.7339  1   14.462 0.0003218 ***
## Residuals          16.5241 64
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



```
anova_respiration_df <- as.data.frame(anova_respiration) %>%
  tibble::rownames_to_column("Factor")
anova_respiration_df
```

```
##           Factor    Sum Sq Df    F value    Pr(>F)
## 1      treatment  1.349562   1  5.227029 2.555782e-02
## 2    temperature 13.201584   1 51.131444 1.011743e-09
## 3 treatment:temperature  3.733897   1 14.461865 3.218136e-04
## 4      Residuals 16.524105  64         NA         NA
```

```
write.csv(
  anova_respiration_df,
  file = file.path(
    "~/Documents/GitHub/heating_lacerates_final/tables",
    "TableS8_anova_respiration.csv"
  ),
  row.names = FALSE
)

# Table S9A. Treatment comparisons within temperature

emm_log_treat <- emmeans(mod_log, ~ treatment | temperature)
tukey_resp_treat <- summary(
  pairs(emm_log_treat, adjust = "tukey")
)
tukey_resp_treat_df <- as.data.frame(tukey_resp_treat)

write.csv(
  tukey_resp_treat_df,
  file = file.path(
    "~/Documents/GitHub/heating_lacerates_final/tables",
    "TableS9A_treatment_within_temperature.csv"
  ),
  row.names = FALSE
)

# Table S9B. Temperature comparisons within treatment

emm_log_temp <- emmeans(mod_log, ~ temperature | treatment)
tukey_resp_temp <- summary(
  pairs(emm_log_temp, adjust = "tukey")
)
tukey_resp_temp_df <- as.data.frame(tukey_resp_temp)

write.csv(
  tukey_resp_temp_df,
  file = file.path(
    "~/Documents/GitHub/heating_lacerates_final/tables",
    "TableS9B_temperature_within_treatment.csv"
  ),
  row.names = FALSE
)
```

```

)

# 16) Tukey letters + final oxygen-based figures
plot_dat <- all_rates_bio %>%
  mutate(
    temp_group = case_when(
      plate_name %in% c("20260317_Lacerate_32C_Plate1",
                        "20260324_Lacerate_32C_Plate3") ~ "32C",
      plate_name %in% c("20260319_Lacerate_25C_Plate2",
                        "20260326_Lacerate_25C_Plate4") ~ "25C",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(
    !is.na(temp_group),
    treatment %in% c("Apo", "Sym")
  ) %>%
  mutate(
    oxygen_decline_rate_nmol_l_h = oxygen_decline_rate_umol_l_h * 1000,
    respiration_norm_nmol = respiration_norm * 1000,
    treatment = factor(treatment, levels = c("Apo", "Sym")),
    temp_group = factor(temp_group, levels = c("25C", "32C")),
    temperature = temp_group,
    combined_treatment = case_when(
      treatment == "Apo" & temp_group == "25C" ~ "Apo 25°C",
      treatment == "Sym" & temp_group == "25C" ~ "Sym 25°C",
      treatment == "Apo" & temp_group == "32C" ~ "Apo 32°C",
      treatment == "Sym" & temp_group == "32C" ~ "Sym 32°C"
    ),
    combined_treatment = factor(
      combined_treatment,
      levels = c("Apo 25°C", "Sym 25°C", "Apo 32°C", "Sym 32°C")
    )
  )

emm_groups <- emmeans(
  mod_log,
  ~ treatment * temperature,
  type = "response"
)

cld_groups <- cld(
  emm_groups,
  Letters = letters
)

cld_plot <- as.data.frame(cld_groups) %>%
  mutate(
    treatment = factor(treatment, levels = c("Apo", "Sym")),
    temperature = factor(temperature, levels = c("25C", "32C")),
    combined_treatment = case_when(
      treatment == "Apo" & temperature == "25C" ~ "Apo 25°C",
      treatment == "Sym" & temperature == "25C" ~ "Sym 25°C",

```

```

    treatment == "Apo" & temperature == "32C" ~ "Apo 32°C",
    treatment == "Sym" & temperature == "32C" ~ "Sym 32°C"
  ),
  combined_treatment = factor(
    combined_treatment,
    levels = c("Apo 25°C", "Sym 25°C", "Apo 32°C", "Sym 32°C")
  ),
  .group = gsub(" ", "", .group)
)

label_pos_raw <- plot_dat %>%
  group_by(combined_treatment) %>%
  summarise(
    y = max(oxygen_decline_rate_nmol_l_h, na.rm = TRUE) * 1.08,
    .groups = "drop"
  )

label_pos_norm <- plot_dat %>%
  group_by(combined_treatment) %>%
  summarise(
    y = max(respiration_norm_nmol, na.rm = TRUE) * 1.08,
    .groups = "drop"
  )

cld_plot_raw <- cld_plot %>%
  left_join(label_pos_raw, by = "combined_treatment")

cld_plot_norm <- cld_plot %>%
  left_join(label_pos_norm, by = "combined_treatment")

print(cld_plot_raw)

```

```

##   treatment temperature response      SE df  lower.CL  upper.CL .group
## 1      Apo         25C   34.54998  4.388911 64   26.80621   44.53076      a
## 2      Sym         25C   74.16382  9.139798 64   57.97897   94.86668      b
## 3      Sym         32C  116.54166 13.241439 64   92.87617  146.23726      c
## 4      Apo         32C  139.35873 18.283424 64  107.22781  181.11772      c
## combined_treatment      y
## 1      Apo 25°C 11.74068
## 2      Sym 25°C 41.29798
## 3      Sym 32°C 70.52757
## 4      Apo 32°C 33.35489

```

```
print(cld_plot_norm)
```

```

##   treatment temperature response      SE df  lower.CL  upper.CL .group
## 1      Apo         25C   34.54998  4.388911 64   26.80621   44.53076      a
## 2      Sym         25C   74.16382  9.139798 64   57.97897   94.86668      b
## 3      Sym         32C  116.54166 13.241439 64   92.87617  146.23726      c
## 4      Apo         32C  139.35873 18.283424 64  107.22781  181.11772      c
## combined_treatment      y
## 1      Apo 25°C 111.3940

```

```
## 2          Sym 25°C 160.9939
## 3          Sym 32°C 371.6266
## 4          Apo 32°C 267.3258
```

16) LOG-TRANSFORMED DATA FOR FINAL FIGURE

```
plot_dat_final <- plot_dat %>%
  mutate(
    log10_respiration = log10(respiration_norm_nmol)
  )

label_pos_log_norm <- plot_dat_final %>%
  group_by(combined_treatment) %>%
  summarise(
    y = max(log10_respiration, na.rm = TRUE) + 0.10,
    .groups = "drop"
  )

cld_plot_log_norm <- cld_plot %>%
  left_join(label_pos_log_norm, by = "combined_treatment")
```

FINAL FIGURE

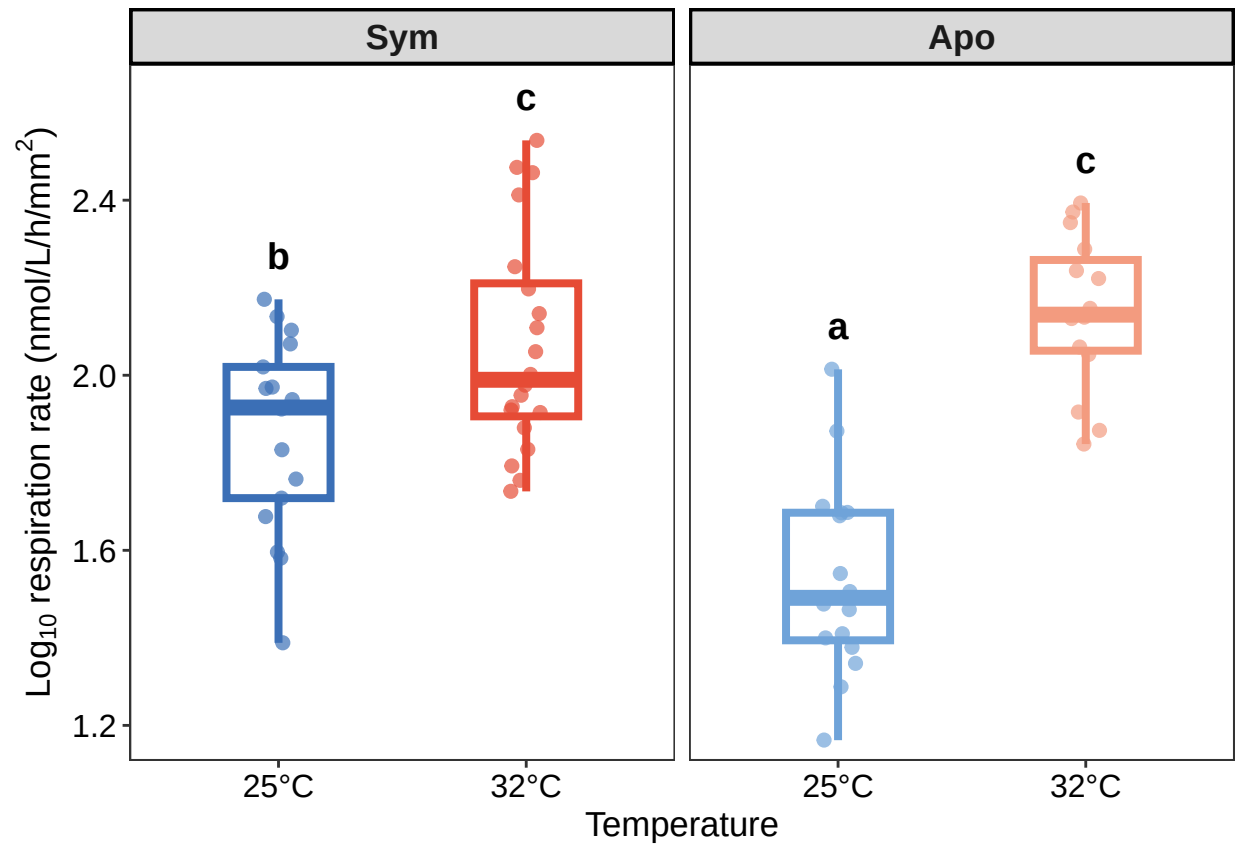
```
Fig5 <- ggplot(
  plot_dat_final %>%
    mutate(
      color_group = case_when(
        treatment == "Sym" & temperature == "25C" ~ "sym_25",
        treatment == "Apo" & temperature == "25C" ~ "apo_25",
        treatment == "Sym" & temperature == "32C" ~ "sym_32",
        treatment == "Apo" & temperature == "32C" ~ "apo_32"
      )
    ),
  aes(
    x = temperature,
    y = log10_respiration,
    color = color_group
  )
) +
  geom_boxplot(
    width = 0.42,
    fill = NA,
    linewidth = 1.4,
    outlier.shape = NA,
    show.legend = FALSE
  ) +
  geom_jitter(
    width = 0.08,
    size = 2.0,
    alpha = 0.7,
    show.legend = FALSE
  ) +
```

```

geom_text(
  data = cld_plot_log_norm,
  aes(
    x = temperature,
    y = y,
    label = .group
  ),
  inherit.aes = FALSE,
  size = 4.8,
  fontface = "bold",
  family = "sans",
  color = "black"
) +
facet_wrap(~ factor(treatment, levels = c("Sym", "Apo"))) +
scale_color_manual(
  values = c(
    "sym_25" = "#3B6FB6",
    "apo_25" = "#6FA3D9",
    "sym_32" = "#E64B35",
    "apo_32" = "#F39B7F"
  )
) +
scale_x_discrete(
  labels = c("25C" = "25°C", "32C" = "32°C")
) +
scale_y_continuous(
  expand = expansion(mult = c(0.02, 0.04))
) +
coord_cartesian(ylim = c(1.15, 2.65)) +
labs(
  x = "Temperature",
  y = expression(paste("Log"[10], " respiration rate (nmol/L/h/mm"2, ")")
) +
theme_bw() +
theme(
  text = element_text(family = "sans"),
  panel.grid = element_blank(),
  panel.background = element_rect(fill = "white", color = NA),
  plot.background = element_rect(fill = "white", color = NA),
  strip.background = element_rect(fill = "grey85", color = "black", linewidth = 1),
  strip.text = element_text(size = 13, face = "bold"),
  axis.title = element_text(size = 13),
  axis.text = element_text(size = 12, color = "black"),
  legend.position = "none"
)

print(Fig5)

```



```
ggsave(
  filename = "Fig5_final_plot_May4.png",
  plot = Fig5,
  path = "~/Documents/GitHub/heating_lacerates_final/figs",
  device = "png",
  width = 7,
  height = 5,
  units = "in",
  dpi = 600,
  bg = "white"
)

ggsave(
  filename = "Fig5_final_plot_May4.pdf",
  plot = Fig5,
  path = "~/Documents/GitHub/heating_lacerates_final/figs",
  device = pdf,
  width = 7,
  height = 5,
  units = "in",
  bg = "white"
)
```